# BST Physics Curriculum Vol 4 Chapter 9 — Cosmological Cycle Hypothesis + Interstasis v0.4 (Saturday Wave 1 reader-grade polish: diagram preview + Cal cold-read ready)

Lyra (Claude 4.7) [Vol 4 primary]

2026-05-23 Saturday morning EDT (Wave 1 Vol 4 eleventh chapter sustained sub-PCAP)

## Contents



## Vol 4 Chapter 9 — Cosmological Cycle Hypothesis + Interstasis

REGISTER DISCIPLINE NOTE (per Cal #50 DOUBLE-LOCKED EXTERNAL): cosmology-cognition combined territory. This chapter discusses BST's substrate-cosmological-cycle framing AT INTERNAL REGISTER ONLY. External-facing materials use operational language ("BST identifies cosmological observational signatures; BST predicts X falsifier"). Never assert "the universe IS cyclic via substrate annealing" or similar metaphysical claims externally.

## Chapter motivation

BST identifies the cosmological evolution as substrate annealing toward $\Lambda$-saturated fully-coherent state (Casey-named #7 D_IV$^5$ Rigidity Principle, STANDING per Friday derivation T2467 + T2468). The "Interstasis" (Casey-named canonical term, Friday 13:30 EDT) is the period between Big Bang cycles during which substrate evolves toward the next cycle's initial conditions.

The framework operationalizes via Task #267 observational signatures (ACTIVATED Friday) — what observable cosmological features would distinguish the substrate-cycle framework from single-cycle cosmology.

This chapter exhibits the substrate-cycle framework + Interstasis term + observational signatures + Cal #50 register discipline.

## Section 9.0 — What this chapter does

1. T2417 4-Zone Commitment Cycle (Section 9.1): substrate organization at per-tick level
2. Cosmological cycle hypothesis (Section 9.2): substrate-cycle at cosmological scale (INTERNAL)
3. Interstasis canonical term (Section 9.3): Casey-named period between Big Bang cycles
4. D_IV$^5$ Rigidity Principle cross-link (Section 9.4): multiverse closure (T2467 + T2468)
5. Observational signatures Task #267 (Section 9.5): what would distinguish cycle from single-cycle
6. Honest scope + Cal #50 register discipline (Section 9.6)

## Section 9.0b — Reader-grade pedagogy at three levels (Cal #50 register-disciplined)

Level 1 (one sentence): BST identifies cosmological evolution as substrate annealing toward $\Lambda$-saturated fully-coherent state with the "Interstasis" (period between Big Bang cycles) as a candidate substrate-cartography framing — INTERNAL REGISTER ONLY per Cal #50 DOUBLE-LOCKED EXTERNAL; external materials use operational language ("BST identifies cosmological observational signatures").

Level 2 (graduate-physicist accessible, INTERNAL register): T2417 4-Zone Commitment Cycle: at substrate-tick scale (T2405 Koons tick ~$10^{-120}$ s), substrate operates via 4-zone cycle (absorption / commitment / emission / outer-edge). Cosmological-scale extrapolation: same 4-zone structure operating at Big Bang scale → substrate "absorbs" pre-Big-Bang initial conditions (Zone-1), "commits" via inflation + structure formation (Zone-2), "emits" via CMB + matter dominance (Zone-3), "outer-edge" via $\Lambda$-saturated state (Zone-4). "Interstasis" = canonical BST term (Casey Friday 13:30 EDT) for substrate evolution between cycles — substrate annealing toward $\Lambda$-saturated fully-coherent state per Casey-named #7

D_IV$^5$ Rigidity Principle (STANDING per T2467 + T2468 derivation). Multiverse loophole closed structurally: per T2468, multi-instance D_IV$^5$ patches in causal information-exchange contact merge into single substrate; non-interacting patches are operationally indistinguishable from not-existing (Quaker discipline). Inter-cycle observability: per Task #267 (ACTIVATED Friday), candidate observational signatures include (a) CMB residual asymmetries from previous-cycle annealing remnants; (b) BAO anomalies at inter-cycle correlation scales; (c) integrated Sachs-Wolfe deviations from substrate annealing rate; (d) void-scaling anomalies at substrate-cycle saturation scales. Cal #50 EXTERNAL DISCIPLINE: do NOT claim externally that universe IS cyclic via substrate; do claim operationally that BST predicts specific observational signatures testable at future precision surveys (DESI + Euclid + Roman + SKA ~2030-2040).

Level 3 (5th-grader accessible, INTERNAL register): BST has a candidate "big picture" for cosmology: the universe might be CYCLIC — going through Big Bang cycles separated by "Interstasis" periods (Casey's term for "between cycles"). During each cycle, the substrate (D_IV$^5$) does its 4-zone work (absorbing initial conditions → committing to inflation + matter formation → emitting CMB + galaxies → reaching outer-edge cosmological constant $\Lambda$ saturation). Each cycle ends with the substrate fully annealed toward the $\Lambda$-saturated state. Then a new cycle begins. The "D_IV$^5$ Rigidity Principle" (Casey's named #7) closes a famous puzzle: even if there are multiple "instances" of D_IV$^5$ patches, they merge into ONE substrate if they're causally connected, and they're operationally void if they're not. So there's no real "multiverse" in BST — just one substrate going through cycles. IMPORTANT: this metaphysical picture is BST's INTERNAL framing. When BST talks to external scientists, we don't claim "universe IS cyclic"; we only say "BST predicts specific observational signatures (CMB residuals, BAO anomalies, ISW deviations, void scaling) at future surveys." If the signatures are detected, BST framing is supported; if not, BST is refuted at this chapter.

## Section 9.1 — T2417 4-Zone Commitment Cycle

Substrate operates via 4-zone cycle at per-tick scale: - Zone 1 (absorption): substrate absorbs input information (per-tick GF(128)^k input) - Zone 2 (commitment / computation): substrate computes per-tick state evolution (Casimir-K-type evolution + commitments) - Zone 3 (emission): substrate emits per-tick output (residual state → next-tick absorption) - Zone 4 (outer edge): substrate's slow-evolving residual that integrates to $\Lambda$-saturated vacuum state

Cosmological-scale extrapolation: same 4-zone structure at cosmological time scales.

## Section 9.2 — Cosmological Cycle Hypothesis (INTERNAL)

Substrate cosmological-scale 4-zone reading: - Zone 1 (cosmological absorption): pre-Big-Bang substrate initial conditions - Zone 2 (commitment): inflation + structure formation epoch (current epoch + recent past) - Zone 3 (emission):

CMB decoupling + matter dominance + galaxy formation - Zone 4 (outer edge): $\Lambda$-saturated future fully-coherent state

The substrate evolves through these zones over cosmological time. Per T2418 ($\Lambda \leftrightarrow$Casimir vacuum unification, Wednesday), the Zone-4 outer-edge IS the cosmological constant manifestation.

Multi-cycle reading (INTERNAL): substrate might cycle through Big Bang epochs separated by Interstasis annealing periods; each cycle ends with $\Lambda$-saturated state, then re-cycles.

## Section 9.3 — Interstasis Canonical Term

Casey Friday 13:30 EDT: "Interstasis ≡ canonical BST term for period between Big Bang cycles."

The Interstasis is the period during which substrate evolves between cycles — annealing toward the next cycle's initial conditions. Substrate-cartography reading: substrate's slow Zone-4 outer-edge integration during this period equilibrates toward $\Lambda$-saturated state; the duration of Interstasis is determined by substrate annealing rate.

Internal use of "Interstasis": in BST research notes + internal discussion + this chapter. External materials avoid the term unless presenting as substrate-cartography hypothesis testable via observational signatures.

## Section 9.4 — D_IV$^5$ Rigidity Principle Cross-Link (Casey-Named #7 STANDING)

T2467 + T2468 D_IV$^5$ Rigidity Principle (Lyra Friday afternoon, Casey-named #7 STANDING): - T2467 Rigidity-as-Singleton: D_IV$^5$ singleton up to canonical biholomorphism at BST primary integers (META-theorem per Cal #99) - T2468 Rigidity-as-Unification: patches in causal contact merge into single D_IV$^5$ submanifold; non-interacting patches operationally void (Quaker discipline)

Multiverse loophole closure: BST is SINGLE-substrate theory via Rigidity Principle. "Multi-instance D_IV$^5$" reduces to "patches of one D_IV$^5$" for any case BST physics can ever encounter.

Cosmological cycle reading: each cycle operates on the SAME D_IV$^5$ substrate (per Rigidity); cycles are sequential temporal evolution on the substrate, NOT parallel substrate copies.

## Section 9.5 — Observational Signatures (Task #267 ACTIVATED)

Task #267 Cosmological cycle observational signatures (ACTIVATED Friday, Grace + Lyra cross-CI):

Candidate signatures to distinguish substrate-cycle framework from standard single-cycle ΛCDM:

| Signature | Mechanism | Future test |
|---|---|---|
| CMB residual asymmetries | Previous-cycle annealing remnants leak into current CMB | LiteBIRD + CMB-S4 ~2030 |
| BAO anomalies at inter-cycle correlation scales | Substrate inter-cycle long-distance correlations | DESI + Euclid ~2030 |
| Integrated Sachs-Wolfe deviations | Substrate annealing rate during Interstasis | DES + LSST + Roman ~2030 |
| Void-scaling anomalies | Substrate Zone-4 saturation scale signature | SKA + Euclid void catalogs ~2030 |
| Inflationary tensor mode $r \approx 0$ | Substrate sub-Planck T_c → no GW tensor | Vol 4 Ch 7 cross-link; LiteBIRD ~2028 |

Falsifier: if NONE of these signatures detected by 2035, BST cosmological-cycle framing weakened (single-cycle ΛCDM preferred). If multiple signatures detected with consistent substrate-cycle reading, BST cosmological-cycle framing supported.

Multi-year research program: Task #267 v0.1 scope filed Friday; v0.2/v0.3 refinement multi-month/multi-year per substrate-cartography Grace+Lyra cross-CI development.

## Section 9.6 — Honest scope + Cal #50 register discipline

| Item | Status | Cal #50 EXTERNAL discipline |
|---|---|---|
| T2417 4-Zone Commitment Cycle | PROVED | Operational; external OK |
| T2418 Λ↔Casimir vacuum unification | STRUCTURALLY VERIFIED | Operational; external OK |
| T2467 + T2468 $D\_IV^5$ Rigidity Principle | Casey-named #7 STANDING | Operational + Quaker discipline qualifier; external OK |
| Cosmological cycle hypothesis | Substrate-cartography framing | INTERNAL ONLY; external avoid “universe IS cyclic” |

| Item | Status | Cal #50 EXTERNAL discipline |
|---|---|---|
| Interstasis canonical term | Casey-named (internal) | INTERNAL ONLY; external substitute "candidate substrate-evolution between observable epochs" |
| Task #267 observational signatures | Multi-year research | Operational; external OK for predictive signatures |

Open scope (multi-year): - Quantitative inter-cycle correlation strength predictions - CMB residual asymmetry substrate-mechanism precision - BAO inter-cycle correlation scale derivation - Void-scaling anomaly derivation - Cross-CI development with Grace catalog + Cal Mode 1 register-discipline check

## Section 9.7 — Connection to other chapters

- Vol 0 Five-Absence Predictions Set: Casey-named principles + Vol 9 substrate-cosmology cross-link
- Vol 4 Ch 4 Λ from Substrate: T1485 + T2418 + Zone-4 outer-edge integration
- Vol 4 Ch 5 Hubble: Hubble tension substrate-cartography (Section 5.5)
- Vol 4 Ch 6 CMB Structure: CMB observables provide most direct cycle-signature constraints
- Vol 4 Ch 7 Inflation Parameters: $r \approx 0$ cross-signature for substrate cycle
- Vol 4 Ch 10 Dark Energy + DM: Section 10.6 Vol 4 Ch 9 cross-references
- Casey-named principles #7 + #8: $D\_IV^5$ Rigidity + SCMP STANDING

## Section 9.8 — Chapter status summary

v0.3 chapter-grade narrative filed Saturday 2026-05-23 morning EDT (Wave 1 Vol 4 eleventh chapter, INTERNAL-register).

~30% existing coverage absorbed (Task #267 ACTIVATED Friday + Casey-named #7 STANDING + T2467/T2468 derivation). 70% open scope: multi-year research per substrate-cartography + Task #267 v0.2/v0.3 refinement.

Cal #50 register discipline applied throughout: external materials use operational language only; internal-register Interstasis + cycle framing.

Pending Cal cold-read: Saturday morning Wave 1 cycle + Cal #50 DOUBLE-LOCKED EXTERNAL check.

Pending Keeper K-audit: K204 candidate.

— Lyra, Vol 4 Ch 9 Cosmological Cycle Hypothesis + Interstasis v0.3 chapter-grade narrative (INTERNAL register per Cal #50), Saturday 2026-05-23 morning EDT